

LPP

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1 Formation of LPP

(using corner points or optimal line)

Example-1: An animal food company receives an order for producing atleast 200kg of a mixture per day, containing two ingredients A and B . The cost of the ingredients A and B is 2 per kg and 8 per kg respectively. A maximum of 80 kg of A and atleast 60 kg of B must be used in preparing the food mixture. Formulate the LP model of the problem to find the amounts of the ingredients used, so that the cost of the food mixture comes to a minimum. Solve it using graphical method.

Solution:- This problem can be considered as a decision-making one. The food company has to take the decision on producing the amount of the ingredients used in preparing the food mixture.

Let x_1 be the amount, in kg, of ingredient A used to produce the food mixture and x_2 be the amount, in kg, of ingredient B . The total amount of food mixture produced is $x_1 + x_2$. Thus we have $x_1 + x_2 \geq 200$. The amount of ingredients A used is not more than 80 kg. Thus, we have $x_1 \leq 80$. Similarly, $x_2 \geq 60$ as atleast 60 kg of ingredient B must be used. Further, the cost involved in producing the food mixture is $2x_1 + 8x_2$. Our objective is to minimize the cost.

Thus, the problem formulated is

$$\begin{aligned} \text{Min } Z &= 2x_1 + 8x_2 \\ \text{subject to, } x_1 + x_2 &\geq 200 \\ x_1 &\leq 80 \\ x_2 &\geq 60 \\ \text{and } x_1, x_2 &\geq 0 \end{aligned}$$

Which is the LP model of formulation.

- Find solution by yourself with graph.

Example-2: A leather company manufactures two types of belts, B_1 and B_2 . Belt B_1 sells at a profit of 3 per belt and Belt B_2 sells at a profit of 2 per belt. Each belt is processed on two machines, M_1 and M_2 . The B_1 type belt requires one minute of processing time on machine M_1 and two minute on machine M_2 . B_2 type belt requires one minute of processing time on each of the machines M_1 and M_2 . While machine M_1 is available for a maximum of 6 h and 40 min, machine M_2 is available for 10 h on all working days. Formulate the given problem as an LPP and solve it using the graphical method.

Solution:- Let x_1, x_2 be the number of belts of type B_1 and B_2 to be manufactured. The LPP formulation of the given problem is as follows:

$$\begin{aligned} \text{Max } Z &= 3x_1 + 2x_2 \\ \text{subject to, } x_1 + x_2 &\leq 400 \\ 2x_1 + x_2 &\leq 600 \\ \text{and } x_1, x_2 &\geq 0 \end{aligned}$$

- Find solution by yourself with graph.

Basic Solutions:- A solution in a L.P.P. which contains m constraints and n variables is said to be basic solution .If

1. atleast $(n - m)$ variable are zero.
2. the m vectors associated to the remaining m variables are Linearly Independent .

$$AX = B$$

$$\Rightarrow X_B = A^{-1}B$$

$$\Rightarrow X_B = [x_{B_1}, x_{B_2}, \dots, x_{B_m}]$$

$$\Rightarrow \text{basic solution : } X_B = [A^{-1}b, 0]$$

Non-degenerate Basic Solution:- If all components of a solution set corresponding to the basic variables are non-zero quantities.

Degenerate Basic Solution:- If some components are zero.

Basic Feasible Solution:- In an L.P.P., a solution satisfying the set of all constraints and the non-negativity conditions is called a feasible solution.

If the feasible solution be basic, then it is called basic feasible solution (B.F.S).

- A B.F.S. contains atleast $(n - m)$ zero variables ,called non-basic variables.
- A B.F.S. contains atleast m positive variables, called basic variables.
- Maximum number of basic solution is atleast $n - m$.where n is number of variables and m is number of constraints.

Variable Unrestricted in Sign:-

The difference of two non-negative variables is a variable unrestricted in sign

$$x_3 = x'_3 - x''_3 \quad x'_3, x''_3 \geq 0.$$

Example:

$$\begin{aligned} & \textbf{Maximize, } Z = 2x_1 + 3x_2 - x_3 \\ & \text{subject to, } 4x_1 + x_2 + x_3 \geq 4 \\ & \quad \quad \quad 7x_1 + 4x_2 - x_3 \leq 25 \\ & \quad \quad \quad x_1, x_3 \geq 0 \text{ and } x_2 \text{ is unrestricted in sign.} \\ \Rightarrow & \textbf{Maximize, } Z = 2x_1 + 3x'_2 - 3x''_2 - x_3 + 0.x_4 + 0.x_5 \\ & \text{subject to, } 4x_1 + x'_2 - x''_2 + x_3 - x_4 = 4 \\ & \quad \quad \quad 7x_1 + 4x'_2 - 4x''_2 - x_3 + x_5 = 25 \\ & \quad \quad \quad x_1, x'_2, x''_2, x_3 \geq 0 \end{aligned}$$

General Form of an LPP:-

$$\begin{aligned} & \text{Optimize } Z = c_1x_1 + c_2x_2 + \dots + c_nx_n, \quad m < n \\ & \text{subject to, } a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n (\leq \text{ or } = \text{ or } \geq) b_1 \\ & \quad \quad \quad a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n (\leq \text{ or } = \text{ or } \geq) b_2 \\ & \quad \quad \quad \vdots \\ & \quad \quad \quad \vdots \\ & \quad \quad \quad a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n (\leq \text{ or } = \text{ or } \geq) b_m \\ & \quad \quad \quad x_1, x_2, \dots, x_n \geq 0 \end{aligned}$$

The matrix form of an LPP is

Optimize $Z = CX$

subject to, $AX (\leq \text{ or } = \text{ or } \geq) b$ and $X \geq 0$.

$$\text{where, } A = \begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{m1} & a_{m2} & \cdot & \cdot & \cdot & a_{mn} \end{bmatrix}_{m \times n}, X = \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix}_{n \times 1}, b = \begin{bmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ b_m \end{bmatrix}_{m \times 1}$$

and $C = (c_1, c_2, \dots, c_n)_{1 \times n}$.

Example: Two linear simultaneous equation with four unknowns are given below -

$$\begin{aligned} 4x_1 + 2x_2 + 3x_3 - 8x_4 &= 6 \\ 3x_1 + 5x_2 + 4x_3 - 6x_4 &= 8 \end{aligned}$$

1. How many basic solutions are there?
2. Find all of them
3. Discuss their nature.

Solution: Here, the equation can be written as

$$a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 = b$$

Where, $a_1 = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$, $a_2 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$, $a_3 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $a_4 = \begin{bmatrix} -8 \\ -6 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$.

1. From four vectors taking two at a time, ${}^4C_2 = 6$ square submatrices are

$$\begin{aligned} B_1 &= (a_1, a_2) = \begin{bmatrix} 4 & 2 \\ 3 & 5 \end{bmatrix}, B_2 = (a_1, a_3) = \begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}, \\ B_3 &= (a_1, a_4) = \begin{bmatrix} 4 & -8 \\ 3 & -6 \end{bmatrix}, B_4 = (a_2, a_3) = \begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix}, \\ B_5 &= (a_2, a_4) = \begin{bmatrix} 2 & -8 \\ 5 & -6 \end{bmatrix}, B_6 = (a_3, a_4) = \begin{bmatrix} 3 & -8 \\ 4 & -6 \end{bmatrix}. \end{aligned}$$

and $\det B_1 = 14 \neq 0$, $\det B_2 = 7 \neq 0$, $\det B_3 = 0$,
 $\det B_4 = -7 \neq 0$, $\det B_5 = 28 \neq 0$, $\det B_6 = 14 \neq 0$.

Here five submatrices are non-singular.

Hence there exist five basis matrices and corresponding to which there exist five basic solutions.

2. For B_1 :

$$B_1^{-1} = \frac{1}{14} \begin{bmatrix} 5 & -2 \\ -3 & 4 \end{bmatrix}$$

$$X_1 = B_1^{-1}b = \frac{1}{14} \begin{bmatrix} 5 & -2 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

the basic solution is $X_1 = [x_1 = 1, x_2 = 1, x_3 = 0, x_4 = 0]$.

For B_2 :

$$B_2^{-1} = \frac{1}{7} \begin{bmatrix} 4 & -3 \\ -3 & 4 \end{bmatrix}$$

$$X_2 = B_2^{-1}b = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

the basic solution is $X_2 = [x_1 = 0, x_2 = 0, x_3 = 2, x_4 = 0]$.

For B_3 :

Not a basis matrix.

$\implies$ no basic solution exist.

$\implies$ x_1 and x_4 together can't be considered basic variables.

For B_4 :

$$X_4 = B_4^{-1}b \implies X_4 = [x_1 = 0, x_2 = 0, x_3 = 2, x_4 = 0].$$

For B_5 :

$$X_5 = B_5^{-1}b \implies X_5 = [x_1 = 0, x_2 = 1, x_3 = 0, x_4 = -\frac{1}{2}] .$$

For B_6 :

$$X_6 = B_6^{-1}b \implies X_6 = [x_1 = 0, x_2 = 0, x_3 = 2, x_4 = 0] .$$

3. X_1 is a non-degenerate B.F.S.
 X_2 is a degenerate B.F.S.
 X_4 is a degenerate B.F.S.
 X_5 is a non-degenerate B.S. but not feasible.
 X_6 is a degenerate B.F.S.

N -Dimensional Euclidean Space :

All n -component row vectors may be considered as points in an n -dimensional space which known as n -dimensional vector space or n -dimensional Euclidean Space denoted by V_n or E^n or $\mathbb{R}^n$ etc.

There are two methods of solving solving a L.P. problem

- Geometrical Method
- Algebraic Method

In algebraic method the problem can be solved only when all constraints are equations. We now show that how the constraints can be converted into equation.

Slack Variables:-

Let some constraints of an L.P.P. are given as

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n \leq b_i, i = 1, \dots, n_1 \in \mathbb{Z}^+.$$

Now, these are transformed into an equation or equality by adding to the left a non-negative variable x_{n+i} . These variables are known as slack variables .

Example:

$$\left. \begin{array}{l} x_1 - 3x_2 + 4x_3 \leq 4 \\ 3x_1 + 2x_2 - x_3 \leq 10 \end{array} \right\} \implies \begin{array}{l} x_1 - 3x_2 + 4x_3 + x_4 = 4 \\ 3x_1 + 2x_2 - x_3 + x_5 = 10 \end{array}$$

Surplus Variables:-

Let some constraints of an L.P.P. be in form

$$a_{k1}x_1 + a_{k2}x_2 + \dots + a_{kn}x_n \geq b_k, k = 1, \dots, n_2 \in \mathbb{Z}^+.$$

Now, these are transformed into an equation or equality by subtracting from the left a non-negative quantity x_{n+k} . These variables are known as surplus variables .

Example:

$$x_1 + 3x_2 + x_3 \geq 12 \implies x_1 + 3x_2 + x_3 - x_6 = 12$$

- If there is a constraint like

$$x_1 - 3x_2 + 7x_3 \geq -20 \implies x_1 - 3x_2 + 7x_3 - x_7 = -20$$

multiply both side by (-1)

$$-x_1 + 3x_2 - 7x_3 + x_7 = 20$$

Here surplus x_7 changes into slack x_7 .

Example: Transform the following L.P.P. problem accordingly such that the simplex method may be applicable. [Express the following minimization problem as standard maximization problem].

$$\begin{aligned} \text{Minimize, } Z &= 4x_1 - x_2 + 2x_3 \\ \text{subject to, } 4x_1 + x_2 - x_3 &\leq 7 \\ 2x_1 - 3x_2 + x_3 &\leq 12 \\ x_1 + x_2 + x_3 &= 8 \\ 4x_1 + 7x_2 - x_3 &\geq 16 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

$\Rightarrow$

$$\begin{aligned} \text{Minimize, } Z &= 4x_1 - x_2 + 2x_3 + 0.x_4 + 0.x_5 + 0.x_6 \\ \text{subject to, } 4x_1 + x_2 - x_3 + x_4 &= 7 \\ 2x_1 - 3x_2 + x_3 + x_5 &= 12 \\ x_1 + x_2 + x_3 &= 8 \\ 4x_1 + 7x_2 - x_3 - x_6 &= 16 \\ x_1, x_2, x_3, x_4, x_5, x_6 &\geq 0 \end{aligned}$$

Here x_1, x_2, x_3 are legitimate variables, x_4, x_5 slack variables and x_6 is surplus variable.

$\Rightarrow$

$$\begin{aligned} \text{Max}(-Z) &= \text{Max}Z^* = -4x_1 + x_2 - 2x_3 + 0.x_4 + 0.x_5 + 0.x_6 \\ \text{subject to, } 4x_1 + x_2 - x_3 + x_4 &= 7 \\ 2x_1 - 3x_2 + x_3 + x_5 &= 12 \\ x_1 + x_2 + x_3 &= 8 \\ 4x_1 + 7x_2 - x_3 - x_6 &= 16 \\ x_1, x_2, x_3, x_4, x_5, x_6 &\geq 0 \end{aligned}$$

Here, $\min Z = -\max(Z^*)$

Note: The two problem will have the same optimal solution set. But actually this not in standard max. form, x_7 is a artificial variable.

- $\boxed{\min(Z) = -\max(-Z)}$, with same set of constraints and same set of solution.

Proof: $z = cx$

Let z attains its min. at $x = x_0$ then

$$\min z = cx_0$$

$$\text{Hence, } cx \geq cx_0$$

$$\text{or, } -cx \leq -cx_0$$

$$\Rightarrow \max(-cx) = -cx_0$$

$$\text{or, } -\max(-cx) = cx_0$$

$$\text{or, } -\max(-z) = \min(z) .$$

1. **Line Segment:** If x_1 and x_2 be two distinct points in E^n , then the line segment joining these two points is a set of points given by

$$X = \{x : x = \lambda x_1 + (1 - \lambda)x_2, 0 \leq \lambda \leq 1, \lambda \in \mathbb{R}\}.$$
2. **Hyperplane:** A set of points in E^n whose co-ordinates satisfy the linear equation of the form $Z = c_1x_1 + c_2x_2 + \dots + c_nx_n$, where $C = (c_1, c_2, \dots, c_n)$, not all $c_i = 0$ and $X = [x_1, x_2, \dots, x_n]^T$, is called a hyperplane for the fixed value z .

$$H = \{x : cx = z\}.$$

Note: A hyperplane $H = \{x : cx = z\}$ in $\mathbb{R}^n$ divides $\mathbb{R}^n$ into three mutually disjoint set as

$$\begin{aligned} X_1 &= \{x : cx < z\} \\ X_2 &= \{x : cx > z\} \\ \text{and } X_3 &= \{x : cx = z\} \end{aligned}$$

- The objective function $Z = CX$ and the constraints with equality sign $Ax = b$ are all hyperplane in a L.P.P.

Example: In $\mathbb{R}^2$, a hyperplane $H = \{x = (x_1, x_2) \in \mathbb{R}^2 : c_1x_1 + c_2x_2 = z\}$ is a straight line.

3. **Convex combination of a set of points:** The convex combination of a set of k points $x_1, x_2, \dots, x_k$ in a space E^n is also a point X in the same space, given by

$$X = \lambda_1x_1 + \lambda_2x_2 + \dots + \lambda_nx_n, \text{ where } \lambda_i \in \mathbb{R} \text{ and } \geq 0 \text{ for all } i$$
and

$$\sum_{i=1}^k \lambda_i = 1$$

Example: If we consider two points x_1, x_2 and x is a convex combination of the points, then

$$x = \lambda_1x_1 + \lambda_2x_2$$

[where $\lambda_1, \lambda_2 \geq 0, \lambda_1 + \lambda_2 = 1 \implies \lambda_2 = 1 - \lambda_1$ or $(1 - \lambda) \geq 0$]

$$= \lambda_1x_1 + (1 - \lambda_1)x_2$$

$$= \lambda x_1 + (1 - \lambda)x_2$$

So, geometrically, the convex combination of two points is a point which lies on the line segment between the points.

4. **Convex Set:** A set X is said to be convex if for any two points $x_1, x_2 \in X$ then the line segment joining the points be also in X .

$$x_1, x_2 \in X \implies x = \lambda x_1 + (1 - \lambda)x_2 \in X \text{ for } 0 \leq \lambda \leq 1.$$
5. **Extreme points of a convex set :** A point x of a convex set X is said to be an extreme point if it can not be expressed as a convex combination of any two other points in X .
6. **Convex Hull:** Let X be a set of points. The set of all convex combinations of the set of points of X is said to be convex hull of X and denoted by $C(X)$.
Example:- i X be the set of points lying on boundary of a circle, then the convex hull of X is the whole circle with its interior.

- The convex hull $C(X)$ of X is the smallest convex set containing X .

7. **Convex Polyhedron:** The set of all convex combination of a finite number of points, is called convex polyhedron of these points.

- Every convex polyhedron is a convex hull but the converse is not true.
- Example:-i, is convex hull but not convex polyhedron.

8. **Simplex:** An n -dimensional convex polyhedron having exactly $(n + 1)$ vertices is called a simplex.

Example:-

1-dim. $\rightarrow$ line segment, 2-dim. $\rightarrow$ triangle, 0-dim. $\rightarrow$ point

9. **Theorem:** A hyperplane is a convex set.

$\implies$ Let $H = \{x : cx = z\}$.

Let $x_1, x_2 \in H$, then $cx_1 = z$ and $cx_2 = z$

Let $x_3 = \lambda x_1 + (1 - \lambda)x_2, 0 \leq \lambda \leq 1$

Now $cx_3 = c[\lambda x_1 + (1 - \lambda)x_2] = \lambda(cx_1) + (1 - \lambda)cx_2 = \lambda z + (1 - \lambda)z = z$

This implies that $x_3 \in H$.

Thus $x_1, x_2 \in H \implies \lambda x_1 + (1 - \lambda)x_2 \in H, 0 \leq \lambda \leq 1$

Hence H is a convex set .

10. The **intersection** of two (finite) convex sets is a convex set .

11. The **union** of two convex set may not be convex set .

12. The set of all feasible solutions (feasible region) is a convex set.

$\implies$ Let us consider an L.P.P.,

$$\begin{array}{l} \text{optimize } Z = CX \\ \text{subject to , } AX = b, X \geq 0 \end{array}$$

Let S be the set of all feasible solutions. Then $S = \{x : Ax = b, x \geq 0\}$

Let $x_1, x_2 \in S$. Then $Ax_1 = b, x_1 \geq 0$ and $Ax_2 = b, x_2 \geq 0$

Let $x_3 = \lambda x_1 + (1 - \lambda)x_2; 0 \leq \lambda \leq 1$

$Ax_3 = A[\lambda x_1 + (1 - \lambda)x_2] = \lambda[Ax_1] + (1 - \lambda)[Ax_2] = \lambda b + (1 - \lambda)b = b$

now $x_3 = \lambda x_1 + (1 - \lambda)x_2 \geq 0$ as $\lambda \geq 0, 1 - \lambda \geq 0, x_1, x_2 \geq 0$

This implies $x_3 \in S$

Hence S is convex .

13. **Reduction of a F.S. to a B.F.S.:**

If a linear programming problem, $AX = b, X \geq 0$, where A is the $m \times n$ co-efficient matrix ($n > m$), $r(A) = m$ has one feasible solution then it has atleast one basic feasible solution.

Example:- If $x_1 = 1, x_2 = 3, x_3 = 2$ is a feasible solution of the equations

$$\begin{array}{l} 2x_1 + 4x_2 - 2x_3 = 10 \\ 10x_1 + 3x_2 + 7x_3 = 33 \end{array}$$

Reduce the above F.S. to a B.F.S. by reduction theorem.

Solution:-

$$A = \begin{bmatrix} 2 & 4 & -2 \\ 10 & 3 & 7 \end{bmatrix} \text{ and } r(A) = 2$$

$\implies a_1, a_2, a_3$ are linearly dependent.

Hence there exist constants $\lambda_1, \lambda_2, \lambda_3$ such that

$\lambda_1 a_1 + \lambda_2 a_2 + \lambda_3 a_3 = 0$ where atleast one of λ_j be non-zero.

$$\implies \lambda_1 \begin{bmatrix} 2 \\ 10 \end{bmatrix} + \lambda_2 \begin{bmatrix} 4 \\ 3 \end{bmatrix} + \lambda_3 \begin{bmatrix} -2 \\ 7 \end{bmatrix} = 0 \quad \dots(1)$$

$$\implies \frac{\lambda_1}{34} = \frac{\lambda_2}{-34} = \frac{\lambda_3}{-34} = k = -\frac{1}{34} \text{ (say)}$$

$$\implies \lambda_1 = -1, \lambda_2 = 1, \lambda_3 = 1$$

$$\text{Hence, } -a_1 + a_2 + a_3 = 0 \quad \dots(2)$$

Taking, $\lambda_1 = -1, \lambda_2 = 1, \lambda_3 = 1$, we have

$$v = \max_j \left(\frac{\lambda_j}{x_j} \right) = \max \left(-\frac{1}{1}, \frac{1}{3}, \frac{1}{2} \right) = \frac{1}{2}$$

Hence a feasible solution given by ,

$$X'_1 = [x_1 - \frac{\lambda_1}{v}, x_2 - \frac{\lambda_2}{v}, x_3 - \frac{\lambda_3}{v}]$$

$$= [1 + 2, 3 - 2, 2 - 2] = [3, 1, 0] \text{ is a B.F.S.}$$

Again (2) can be written as

$$a_1 - a_2 - a_3 = 0$$

$$\text{then } \lambda_1 = 1, \lambda_2 = -1, \lambda_3 = -1$$

$$v = \max \left(\frac{1}{1}, -\frac{1}{3}, -\frac{1}{2} \right) = 1$$

Hence a feasible solution given by-

$$X'_2 = [1 - 1, 3 + 1, 2 + 1] = [0, 4, 3]$$

Note:-

If all λ_i are non-negative or non-positive only one non-degenerate B.F.S. can be obtained by using the theory.

2 Simplex Method

We shall now try to systemize the previous ideas into a computational process suitable for numerical solution of the problems which is known as the simplex algorithm .

First, we take an example to explain the procedure:-

2.1 Example 1

Solve the following L.P.P.:

$$\begin{aligned} \text{Max } Z &= x_1 + x_2 + 3x_3 \\ \text{subject to, } 3x_1 + 2x_2 + x_3 &\leq 3 \\ 2x_1 + x_2 + 2x_3 &\leq 2 \\ \text{and } x_1, x_2, x_3 &\geq 0. \end{aligned}$$

Solution:-

In standard form after introducing slack variables x_4 and x_5

$$\begin{aligned} \text{max } z &= x_1 + x_2 + 3x_3 + 0.x_4 + 0.x_5 \\ \text{subject to } 3x_1 + 2x_2 + x_3 + x_4 &= 3 \\ 2x_1 + x_2 + 2x_3 + x_5 &= 2 \end{aligned}$$

Here, a_4 and a_5 gives the initial basis matrix , as such x_4 and x_5 are the initial basic variables. Now, $B = I_2$ is the initial admissible basis to start the problem.

$$\begin{aligned} X_B &= [x_{B_1}, x_{B_2}] = [x_4, x_5] = [b_1, b_2] = [3, 2] \\ C_B &= (C_{B_1}, C_{B_2}) = (C_4, C_5) = (0, 0) \\ Z_B &= C_B X_B = 0 \\ y_{ij} &= a_{ij} \end{aligned}$$

Initial Simplex Table

			C_j	1	1	3	0	0	
C_B	Basis	X_B	b	a_1	a_2	a_3	a_4	a_5	Min Ratio $\frac{x_{B_i}}{y_{i3}}, y_{i3} > 0$
0	a_4	x_4	3	3	2	1	1	0	$\frac{3}{1} = 3$
0	a_5	x_5	2	2	1	2	0	1	$\frac{2}{2} = 1$ (Min)
$Z_j - C_j$				-1	-1	-3↑	0	0↓	

$$Z_j - C_j = C_B \cdot y_j - C_j$$

For entering vector:-

Now, $\min\{Z_j - C_j; Z_j - C_j < 0\} = -3 \implies a_3$ is the **entering vector**.

For departing vector:-

$\min\{\frac{x_{B_i}}{y_{ik}} | y_{ik} > 0\} \min\{\frac{3}{1}, \frac{2}{2}\} = 1 \implies a_5$ is the **departing vector**.

For the presence of $Z_j - C_j < 0$, we proceed to the next table.

2nd simplex table (1st iteration)

			C_j	1	1	3	0	0	
C_B	Basis	X_B	b	a_1	a_2	a_3	a_4	a_5	Min Ratio
0	a_4	x_4	2	2	$\frac{3}{2}$	0	1	$-\frac{1}{2}$	
3	a_3	x_3	1	1	$\frac{1}{2}$	1	0	$\frac{1}{2}$	
$Z_j - C_j$				2	$\frac{1}{2}$	0	0	$\frac{3}{2}$	

Do operations, $R'_2 = R_2/2$ and $R'_1 = R_1 - R_2$

The solution is optimal as $Z_j - C_j \geq 0 \quad \forall j$

The optimal solution is unique as for all non-basic vectors $Z_j - C_j > 0$ and $Z_j - C_j = 0$ for all basic vectors .

Hence the **required solution is**

$x_1 = 0, x_2 = 0, x_3 = 1$ and $Z_{max} = 3$.

Notes:

1. If all $Z_j - C_j \geq 0 \forall j$ then the present B.F.S is optimal .(for maximization problem)
2. If, for some a_j (at least one j), $Z_j - C_j < 0$ and $y_{ij} \leq 0$ for all j , then the given problem will have an unbounded solution and leave it here.
3. If none of $Z_j - C_j < 0$ but some $Z_j - C_j = 0$ corresponding to non-basic vectors, multiple solutions exist.

Example: (Problem having Multiple Optimal solutions)

Use the simplex method to solve the following L.P.P.

$$\begin{aligned} &\textbf{Maximize} , Z = 5x_1 + 2x_2 \\ &\text{subject to, } 6x_1 + 10x_2 \leq 30 \\ &\quad 10x_1 + 4x_2 \leq 20 \\ &\quad \text{and } x_1, x_2 \geq 0. \end{aligned}$$

Is the solution unique? If not, write down the convex combination of the alternative optima.

Solution:-

The constraints, after addition of slack variables x_3 and x_4 , one to each are

$$\begin{aligned} 6x_1 + 10x_2 + x_3 &= 30 \\ 10x_1 + 4x_2 + x_4 &= 20, x_j \geq 0, j = 1, 2, \dots, 4. \end{aligned}$$

The adjusted objective function, $\mathbf{Z} = 5\mathbf{x}_1 + 2\mathbf{x}_2 + 0\mathbf{x}_3 + 0\mathbf{x}_4$.

$b = \begin{bmatrix} 30 \\ 20 \end{bmatrix} \geq 0$ and $B = (a_3, a_4) = I_2$ is a unit matrix .

$$X_B = B^{-1}b = b \geq 0.$$

Thus, with the initial basis B , we can start the problem

$$y_j = B^{-1}a_j = a_j, c_B = (0, 0).$$

		C_j	5	2	0	0	
C_B	Basis	b	a_1	a_2	a_3	a_4	Min Ratio
0	a_3	30	6	10	1	0	$\frac{30}{6} = 5$
0	a_4	20	10	4	0	1	$\frac{20}{10} = 2$
$Z_j - C_j$		0	-5	-2	0	0	
0	a_3	18	0	$\frac{38}{5}$ *	1	$-\frac{3}{5}$	$\frac{18}{\frac{38}{5}} = \frac{90}{38}$
5	a_1	2	1	$\frac{2}{5}$	0	$\frac{1}{10}$	$\frac{2}{\frac{2}{5}} = 5$
$Z_j - C_j$		10	0	0	0	$\frac{1}{2}$	
2	a_2	$\frac{45}{19}$	0	1	$\frac{5}{38}$	$-\frac{3}{38}$	
5	a_1	$\frac{20}{19}$	1	0	$-\frac{2}{19}$	$\frac{5}{38}$	
$Z_j - C_j$		10	0	0	0	$\frac{1}{2}$	

Here in the second table all $Z_j - C_j \geq 0$.

Hence the optimal solution has been obtained $\max Z = 10$ at $x_3 = 18, x_1 = 2$, i.e., for $x_1 = 2, x_2 = 0$ (non-basic), the problem attains its maximum.

But here $z_2 - c_2 = 0$ corresponding to a non-basic vector a_2 .

Thus the solution is not unique .Using a_2 as a vector to enter in the next basis we have $\max Z = 10$ remains same but the optimal solution will change which has been shown from the table 3 .Other optimal basic solution is $x_1 = \frac{20}{19}, x_2 = \frac{45}{19}$.

We know that if there are more than one optimal solution then there exist infinite optimal solutions which will be obtained from the convex combination of the optimal solutions $x_1 = [2, 0], x_2 = [\frac{20}{19}, \frac{45}{19}]$. Any optimal solution x is given by **[Alternative optima]**.

$$\begin{aligned} X &= \lambda X_1 + (1 - \lambda) X_2, 0 \leq \lambda \leq 1 \\ &= \lambda [2, 0] + (1 - \lambda) [\frac{20}{19}, \frac{45}{19}] \end{aligned}$$

For example, if we take $\lambda = \frac{1}{2}$ then $x = [1\frac{10}{19}, \frac{45}{38}]$ which is also an alternative optimal solution.

Note: $X_1 = [x_1 = 2, x_2 = 0]$

Example : (Problem having an Unbounded Solution)

Use the simplex method to solve the following L.P.P.:

$$\begin{aligned} &\textbf{Maximize, } 2x_2 + x_3 \\ &\text{subject to, } x_1 + x_2 - 2x_3 \leq 7 \\ &\quad -3x_1 + x_2 + 2x_3 \leq 3 \\ &\text{and } x_1, x_2, x_3 \geq 0. \end{aligned}$$

Solution:-

The constraints ,after addition of slack variables x_4 and x_5 ,one to each are

$$\begin{aligned} x_1 + x_2 - 2x_3 + x_4 &= 7 \\ -3x_1 + x_2 + 2x_3 + x_5 &= 3, x_j \geq 0, j = 1, 2, \dots, 5 \end{aligned}$$

and the objective function is $0.x_1 + 2x_2 + x_3 + 0.x_4 + 0.x_5$, $b = [7, 3] \geq 0$ and $B = [a_4, a_5] = I_2$ will be the initial unit basis.

		C_j	0	2	1	0	0	
C_B	Basis	b	a_1	a_2	a_3	a_4	a_5	Min Ratio
0	a_4	7	1	1	-2	1	0	$\frac{7}{1} = 7$
0	a_5	3	-3	1	2	0	1	$\frac{3}{1} = \mathbf{3}$
$Z_j - C_j$		0	0	-2 ↑	-1	0	0	
0	a_4	4	4	0	-4	1	-1	$\frac{4}{4} = \mathbf{1}$
2	a_2	3	-3	1	2	0	1	
$Z_j - C_j$		6	-6 ↑	0	3	0	2	
0	a_1	1	1	0	-1	$\frac{1}{4}$	$-\frac{1}{4}$	
2	a_2	6	0	1	-1	$\frac{3}{4}$	$\frac{1}{4}$	
$Z_j - C_j$		12	0	0	-3	$\frac{3}{2}$	$\frac{1}{2}$	

Note:-

1. After completing the $z_j - c_j$ of the third table, we have seen that $z_3 - c_3 = -3 < 0$. Thus we require to complete the table. After completing, we have seen that $y_{i3} \leq 0$ for $i = 1, 2$ for which $z_3 - c_3$ is negative. then the only conclusion is that the problem has no finite optimal value, and the problem is said to have an unbounded solution.
2. For a problem having an unbounded solution, we cannot trace it before completing the final table.

Example(Transfer Minimization to Maximization Problem)

Use the simplex method to solve the following L.P.P.:

$$\begin{aligned} \text{Minimize, } Z &= -2x_1 + 3x_2 \\ \text{subject to, } 2x_1 - 5x_2 &\leq 7 \\ 4x_1 + x_2 &\leq 8 \\ 7x_1 + 2x_2 &\leq 16 \\ \text{and } x_1, x_2 &\geq 0. \end{aligned}$$

Solution:-

The problem is a problem of minimization.

Let $z' = -z$; then $\min z = -\max(-z) = -\max z'$.

Hence the problem is a problem of maximization of $z' = -z = -(-2x_1 + 3x_2) = 2x_1 - 3x_2$ and finally $\min z = -\max z'$ with the same solution set.

$b_i \geq 0$ for all i and constraints are associated with the sign “ $\leq$ ”.

Introducing three slack variables x_3, x_4 and x_5 (one in each inequation) we get the following converted equations

$$\begin{aligned} 2x_1 - 5x_2 + x_3 &= 7 \\ 4x_1 + x_2 + x_4 &= 8 \\ 7x_1 + 2x_2 + x_5 &= 16. \end{aligned}$$

The adjusted objective function is $z' = 2x_1 - 3x_2 + 0.x_3 + 0.x_4 + 0.x_5$

Here all the slack vectors are unit vectors which produce a unit basis .Initial

$$B.F.S = X_B = [x_{B1}, x_{B2}, x_{B3}] = [x_3, x_4, x_5] = [7, 8, 16]$$

$$C_B = (C_{B1}, C_{B2}, C_{B3}) = (c_3, c_4, c_5) = (0, 0, 0)$$

$$Z = C_B X_B = 0 \text{ and } y_j = B^{-1}a_j = a_j$$

$$Z_j - C_j = C_B \cdot y_j - C_j = 0 \cdot y_j - C_j = -C_j$$

Now with the values of $Z_j - C_j$ etc. we construct the initial table and solve accordingly.

		C_j	2	-3	0	0	0	
C_B	Basis	b	a_1	a_2	$a_3(e_1)$	$a_4(e_2)$	$a_5(e_3)$	Min Ratio
0	a_3	7	2	-5	1	0	0	$\frac{7}{2}$
0	a_4^*	8	4	1	0	1	0	$\frac{8}{4} = 2$
0	a_5	16	7	2	0	0	1	$\frac{16}{7}$
$Z_j - C_j$		0	-2↑	3	0	0	0	
0	a_3	3	0	$-\frac{11}{2}$	1	$-\frac{1}{2}$	0	
2	a_1	2	1	$\frac{1}{4}$	0	$\frac{1}{4}$	0	
0	a_5	2	0	$\frac{1}{4}$	0	$-\frac{7}{4}$	1	
$Z_j - C_j$		4	0	$\frac{7}{2}$	0	$\frac{1}{2}$	0	

As none of $Z_j - C_j < 0$, therefore the solution is optimal.

Hence $\max z' = 4$.

Now $\min z = -\max z' = -4$. Hence the minimum value of Z is -4 corresponding to the optimal basic feasible solution.

$X_B = [x_3, x_1, x_5] = [3, 2, 2]$, i.e., for $x_1 = 2, x_2 = 0$, the objective function of the original problem attains its minimum [x_2 is a non-basic variable.].

Example: Solve the L.P. problem by simplex method.

$$\begin{aligned} \text{Maximize, } z &= 3x_1 + 5x_2 + 4x_3 \\ \text{subject to, } 2x_1 + 3x_2 &\leq 8 \\ 2x_2 + 5x_3 &\leq 10 \\ 3x_1 + 2x_2 + 4x_3 &\leq 15 \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

Solution:- $b = [8, 10, 15] \geq 0$. Thus introducing three slack variables, x_4, x_5 and x_6 , one to each constraint and taking initial basis $B = (a_4, a_5, a_6) = I_3$, we can start the initial simplex table and then solve in a compact table as shown below.

		C_j	3	5	4	0	0	0	
C_B	basis	b	a_1	a_2	a_3	$a_4(e_1)$	$a_5(e_2)$	$a_6(e_3)$	Min. ratio
0	a_4^*	8	2	3	0	1	0	0	$\frac{8}{3} = 2\frac{2}{3}$
0	a_5	10	0	2	5	0	1	0	$\frac{10}{2} = 5$
0	a_6	15	3	2	4	0	0	1	$\frac{15}{2}$
$Z_j - C_j$		0	-3	-5 $\uparrow$	-4	0	0	0	
5	a_2	$\frac{8}{3}$	$\frac{2}{3}$	1	0	$\frac{1}{3}$	0	0	...
0	a_5^*	$\frac{14}{3}$	$-\frac{4}{3}$	0	5	$-\frac{2}{3}$	1	0	$\frac{14}{5} = 2\frac{2}{5}$
0	a_6	$\frac{29}{3}$	$\frac{5}{3}$	0	4	$-\frac{2}{3}$	0	1	$\frac{29}{4} = 7\frac{1}{4}$
$Z_j - C_j$		$\frac{40}{3}$	$\frac{1}{3}$	0	-4 $\uparrow$	$\frac{5}{3}$	0	0	
5	a_2	$\frac{8}{3}$	$\frac{2}{3}$	1	0	$\frac{1}{3}$	0	0	$\frac{8}{3} = 2\frac{2}{3}$
4	a_3	$\frac{14}{15}$	$-\frac{4}{15}$	0	1	$-\frac{2}{15}$	$\frac{1}{5}$	0	...
0	a_6^*	$\frac{89}{15}$	$\frac{41}{15}$	0	0	$-\frac{2}{25}$	$-\frac{4}{5}$	1	$\frac{89}{15} = 5\frac{14}{15}$
$Z_j - C_j$		$\frac{256}{15}$	$-\frac{11}{15} \uparrow$	0	0	$\frac{17}{15}$	$\frac{4}{5}$	0	
5	a_2	$\frac{50}{41}$							
4	a_3	$\frac{62}{41}$							
3	a_1	$\frac{89}{41}$							
$Z_j - C_j$		$\frac{765}{41}$	0	0	0	$\frac{45}{41}$	$\frac{24}{41}$	$\frac{11}{41}$	

In the fourth table, all $z_j - c_j \geq 0$. Thus the fourth table is the optimal table. We now only calculate the elements under the column vector b which gives B.F.S. and the value of the objective function. We need not complete the table.

$$\max z = \frac{765}{41} \text{ at } x_1 = \frac{89}{41}, x_2 = \frac{50}{41} \text{ and } x_3 = \frac{62}{41}.$$

Example: Solve the L.P.P. by simplex method.

$$\begin{aligned} &\text{Minimize, } z = x_1 - 3x_2 + 2x_3 \\ &\text{subject to, } 3x_1 - x_2 + 2x_3 \leq 7 \\ &\quad \quad \quad -2x_1 + 4x_2 \leq 12 \\ &\quad \quad \quad -4x_1 + 3x_2 + 8x_3 \leq 10 \\ &\quad \quad \quad x_1, x_2, x_3 \geq 0. \end{aligned}$$

Solution:-

		C_j	1	-3	2	0	0	0	
C_B	Basis	b	a_1	a_2	a_3	$a_4(e_1)$	$a_5(e_2)$	$a_6(e_3)$	Min. ratio
0	a_4	7	3	-1	2	1	0	0	
0	a_5^*	12	-2	4	0	0	1	0	$\frac{12}{4} = 3$
0	a_6	10	-4	3	8	0	0	1	$\frac{10}{3}$
$Z_j - C_j$		0	-1	3 ↑	-2	0	0	0	
0	a_4^*	10	$\frac{5}{2}$	0	2	1	$\frac{1}{4}$	0	$\frac{10}{\frac{5}{2}} = 4$
-3	a_2	3	$-\frac{1}{2}$	1	0	0	$\frac{1}{4}$	0	
0	a_6	1	$-\frac{5}{2}$	0	8	0	$-\frac{3}{4}$	1	
$Z_j - C_j$		-9	$\frac{1}{2} \uparrow$	0	-2	0	$-\frac{3}{4}$	0	
1	a_1	4							
-3	a_2	5							
0	a_6	11							
$Z_j - C_j$		-11	0	0	$-\frac{12}{5}$	$-\frac{1}{5}$	$-\frac{4}{5}$	0	

All $z_j - c_j \leq 0$. Thus we reach at the optimal stage.

$\min(z) = -11$ at $x_1 = 4, x_2 = 5$ and $x_3 = 0$ (non-basic).